Venkata Rajesh Paidipala

EMAIL: pvr12116@gmail.com **PH: 513-394-3009** **OH, Cincinnati (willing to relocate)**

PROFESSIONAL SUMMARY:

- Network Engineer with hands-on experience in routing, switching, and **securing enterprise** and service provider networks using protocols like **OSPF, BGP, MPLS, VLANs, and VPNs**.
- Proficient in **network security and firewall configurations** (Cisco ASA, Palo Alto), implementing VPNs, ACLs, and IDS/IPS for secure data transmission and threat prevention.
- **Skilled in cloud networking** on **AWS and Azure**, including **VPC setup**, **IAM** policy management, and hybrid cloud integration for scalable infrastructure.
- Experienced in **automating network tasks using Python, Perl, Ansible, and Terraform**, improving deployment speed, consistency, and reducing manual errors.
- Adept at using tools such as **Wireshark, PRTG, and SolarWinds for real-time monitoring**, traffic analysis, and performance optimization.
- **Strong team player** with a proactive approach **to problem-solving**, committed to building secure, high-performing, and resilient network systems.

TECHNICAL SKILLS:

Core Networking:

OSPF, BGP, EIGRP, RIP, MPLS, **VLANs, VTP, STP/RSTP, EtherChannel, QoS, VRF,** Segment Routing, DNS, DHCP, NTP, STP, RSTP, EtherChannel, NAT, SNMP, IP/MPLS, L2/L3 VPNs, CLOS architecture.

Security:

Cisco ASA, Palo Alto, FortiGate**, Checkpoint Firewalls, IDS/IPS, VPNs (IPSec/OpenVPN), TACACS+**, RADIUS, **Zero Trust Architecture, ACLs**, SIEM tools, Vulnerability Management

Cloud & Virtualization:

AWS (**VPC, EC2, S3, IAM, Route53, CloudFormation**), Azure Networking, **GCP Networking**, VMware, Hyper-V.

Automation & Scripting:

Python, Perl, Ansible, Terraform, **Bash, PowerShell**, YAML, JSON

DevOps & Tools:

Docker, Kubernetes, Jenkins, **GitHub Actions, GitLab CI, Git**, Confluence, Agile/Scrum

Monitoring & Testing Tools:

Wireshark, SolarWinds, PRTG, **NetFlow, Syslog, Spirent**, IXIA, Cisco Packet Tracer, Nmap, Metasploit

OperatingSystems:

Linux (**Ubuntu, CentOS, Red Hat, Debian**), Shell Scripting

Experience:

Company: Tejas Networks **Sep 2021 – Dec 2023**

Role: Research and Development PE Engineer

Technologies: MPLS, OSPF, BGP, RSVP-TE, L2VPN, IPV4,IPV6, SD-WAN, VLANs, Python, Linux, Ansible, NMS/EMS, Spirent, IXIA

Responsibilities:

- Conducted detailed manual and automated testing of advanced routing protocols (**OSPFv3, BGP, EIGRP, RSVP-TE**) across multi-vendor environments including **Cisco, Juniper, and Huawei.**
- Validated L2/L3 VPNs, VPLS, and **MPLS** configurations for secure and scalable deployments in enterprise and **SD-WAN** setups.
- Integrated Tejas equipment into existing service provider networks, coordinating interoperability with **Nokia, Aruba, Juniper, and Cisco platforms**.
- Managed **VLANs, VPNs, firewall** configurations (Cisco ASA, Palo Alto), and network monitoring via **EMS/NMS** tools.
- Participated in **Vulnerability Assessment and Penetration Testing** (**VAPT**) for the TJ1400P-H switch, supporting security compliance.
- Automated test scenarios and configurations using **Python and Ansible**, improving testing **efficiency and accuracy**.
- Conducted DWDM testing on TJ1600 series for Power Grid deployment, ensuring signal quality over 500+ km links.

Company: Capgemini **Nov 2020 – August 2021**

Role: Network Analyst

Technologies:

OSPF, BGP, VLANs, QoS, Wireshark, PRTG, SolarWinds, SNMP, Python, PowerShell

Responsibilities:

- Configured **L2 switches** and routing protocols (**OSPF, BGP**) to optimize enterprise network performance.
- Analyzed **real-time traffic using Wireshark and hping** to detect anomalies and evaluate **DDoS/DoS** resilience.
- Applied **QoS strategies** to ensure consistent delivery of business-critical services.
- Implemented and enforced **network security protocols**, monitoring **SNMP traps** and setting up alerts for incident response.
- Delivered rapid troubleshooting support in production environments, **minimizing downtime** and ensuring uptime targets.
- Developed custom scripts in **Python and PowerShell for routine automation** and network diagnostics.

Education:

Master of Science in Information Technology **Jan 2024 – April 2025**

University of Cincinnati **GPA : 3.93 Scale of 4.0**

Projects:

1. **Title: Comprehensive Network Optimization and Secure SD-WAN Deployment. -Tejas Netw**

Technologies: OSPF, BGP, VLANs, Cisco IOS, ACLs**, MPLS, SD-WAN, VPN, QoS, Network Security, Wireshark,** Network Monitoring Tools**.**

Description:

- Designed and deployed dynamic routing (OSPF, RIP, EIGRP, BGP) for a **robust and redundant infrastructure**.
- Segmented **network traffic** using **VLANs** and implemented **ACLs** for access control.
- Established encrypted **VPN tunnels** and deployed SD-WAN to improve **multi-site connectivity**.
- Prioritized VoIP and video traffic with **QoS** and analyzed performance using Wireshark and other tools.

2. **Title: Zero Trust Security Architecture on AWS - Univ of Cincinnati**

Technologies Used: AWS IAM, VPC, Security Groups, NACLs, AWS WAF, AWS SSO, CloudTrail, Guard Duty, Terraform.

Description:

- Built a Zero Trust model on **AWS with strict access policies, MFA,** and segmented networks.
- Applied **VPC designs** with secure subnetting**, WAF rules**, and centralized logging via **CloudTrail.**
- Deployed Guard Duty for threat detection and automated secure infrastructure provisioning with Terraform.

3. **Secure VPC Architecture with Public & Private Subnets and Bastion Host - Univ of Cincinnati**

Technologies Used: AWS VPC, EC2, Subnets, Internet Gateway, Route Tables, **Security Groups,** SSH**,** Linux

- Built a **secure AWS VPC with public and private subnets** to isolate internal servers and minimize external exposure.
- Configured a Bastion Host in the public subnet to enable **secure SSH access to private EC2 instances** using role-based restrictions.
- Managed routing and **firewall rules** through **custom route tables and security groups**, ensuring granular control over **traffic flow.**
- Applied **real-world security practices** including **least privilege access, subnet segmentation, and jump-box architecture** for enhanced protection.

4. **Cybersecurity Mini-Project - Univ of Cincinnati**

Tools/Technologies: Python, PyCharm, Cipher Text, Morse Code, Basic Cryptography

- Designed a Python-based application to **encode and decode messages** using classic cryptographic methods including **Morse code** and **Caesar Cipher**.
- Gained insight into **how attackers exploit weak encryption** and how to strengthen it using layered encoding and secure key exchanges.
- Practiced secure development using **PyCharm**, including input validation and exception handling to prevent misuse.